

CLINICAL FINDINGS

Staphylococcal Toxic Shock Syndrome

The most common staphylococcal toxin associated with toxic shock syndrome (TSS) is **TSS toxin-1 (TSST-1)**, which is the predominant toxin in menstrual-associated cases. TSST-1 is unique among superantigens in its ability to cross mucosal surfaces. Following the removal of highly absorbent tampons from the market and increased patient education, the incidence of menstruation-associated TSS has declined steadily. Currently, **non-menstrual TSS exceeds menstrual-associated cases** in frequency.

Non-menstrual TSS is associated with: - Postsurgical wounds - Sinusitis - Osteomyelitis - Influenza - Intravenous drug use - Burn wounds - Gynecologic infections (especially in the postpartum period)

Other staphylococcal toxins, including **staphylococcal enterotoxins B and C (SEB and SEC)**, are also implicated. SEB and SEC account for approximately **50% of non-menstrual TSS cases**. Host susceptibility plays a key role, as individuals lacking neutralizing antibodies against TSST-1 are at increased risk.

Clinical Presentation of Staphylococcal TSS

Symptoms begin with the acute onset of: - Fever - Sore throat - Myalgia

Gastrointestinal symptoms are common: - Diarrhea - Vomiting

The rash is typically a **macular erythema**, though it may have a **scarlatiniform** appearance. It usually starts on the trunk and spreads to the extremities, often involving the **palms and soles**. In hypotensive patients, the rash tends to be more prominent on the trunk. Hypotension may manifest as orthostatic dizziness, syncope, or overt shock.

Key clinical signs include: - **Nonpurulent conjunctival hyperemia** - **Pharyngeal inflammation** - **Strawberry tongue** (similar to scarlet fever) - Altered mental status

The rash typically **desquamates 1 to 2 weeks** after onset, particularly on the palms and soles.

The **Centers for Disease Control and Prevention (CDC)** criteria for TSS are used for diagnosis. Milder cases that do not meet full diagnostic criteria may occur, especially with early recognition and treatment. In non-menstrual cases—particularly postoperative infections—classic local signs of infection (e.g., erythema, pain, purulence) may be absent, which contrasts with streptococcal TSS.

Streptococcal Toxic Shock Syndrome

Streptococcal TSS was first described in the late 1980s as a clinical entity similar to staphylococcal TSS but caused by **invasive group A streptococcus (GAS)**. Recent evidence suggests that **streptococcal TSS may be more common** than the staphylococcal form.

The majority of cases are mediated by **streptococcal pyrogenic exotoxin A (SPEA)**, a superantigen. Other superantigenic toxins involved include: - **SPEB** - **SPEC**

Cases involving **non-group A streptococci** have also been reported.